BUSA 390B: Marketing Analytics

Assignment 2: DoorDash

Student Name: Vy (Valerie) Mai Le

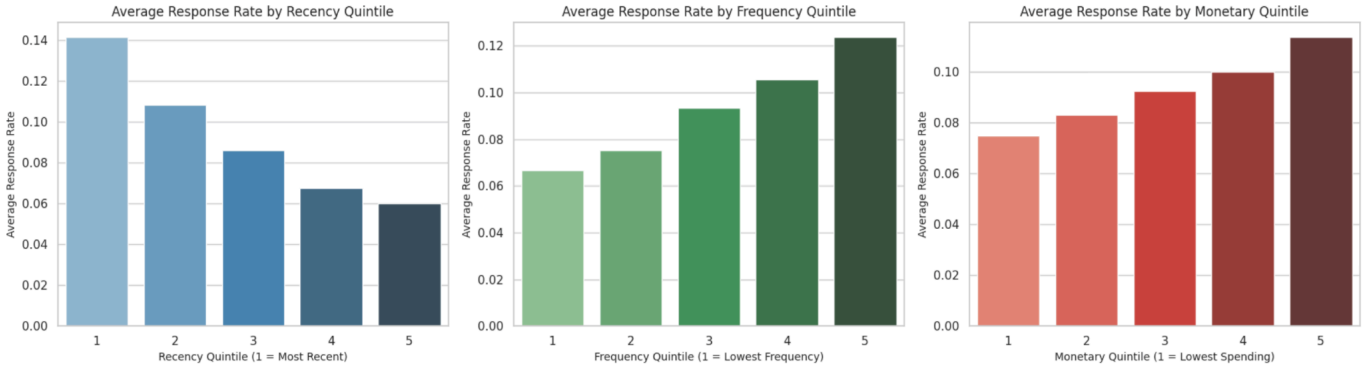
Student Number: 932709

Question 1: Average Response Rate and Profit per Order

From the dataset, the **average response rate** to the promotional offer is **9.29%**, meaning that about one in every eleven customers redeemed the offer within the campaign period. Among those who responded, DoorDash earned an **average profit of $9.90 per order** after accounting for all delivery, restaurant, and operational costs.

This suggests that, although most customers did not redeem the promotion, the customers who did generated a reasonable level of profit per transaction, indicating that the offer did motivate profitable ordering behavior among a subset of users.

Question 2: RFM Quintiles and Response Patterns



When dividing customers into quintiles based on **Recency, Frequency, and Monetary (RFM)** values, three clear trends emerge:

- **Recency:** Customers who ordered more recently (lower recency scores) showed the highest response rates. This confirms that recency is a strong indicator of engagement: recent customers are more responsive to promotions.

- **Frequency:** Response rates rise steadily across frequency quintiles, suggesting that customers who order more frequently are more likely to respond to push notifications.
- **Monetary:** Higher-spending customers also exhibit slightly higher response rates, though the effect is weaker than for recency and frequency.

From a managerial perspective, this implies that **recency and frequency** should be prioritized when designing targeted campaigns. Recently active and frequent users are both more responsive and more likely to generate incremental sales, making them ideal segments for selective promotion.

Question 3: Breakeven Response Rate

Using the marginal cost of sending a push notification (**$0.96**) and the average profit per order (**$9.90**), the **breakeven response rate** is:

$$\text{Breakeven Response Rate} = \text{Cost per promotion} / \text{Profit per order}$$

or

$$\text{Breakeven Response Rate} = 0.96 / 9.90 \approx 0.097 \ (9.7\%)$$

This means that for DoorDash to avoid losing money, at least **9.7% of targeted customers** must respond to the promotion. Any campaign achieving a response rate below this threshold would result in negative net profit.

Question 4: Sending the Promotion to Everyone

If DoorDash were to send the push notification to all remaining **36,930,000 customers**, the results would be as follows:

- **Expected responders:** 3,430,269
- **Total projected profit from orders:** $33,955,969.07
- **Marketing spend:** $35,452,800.00
- **Net projected profit: –$1,496,830.93**
- **Return on Marketing Investment (ROMI): –0.04 (-4%)**

In this "send-to-all" strategy, DoorDash would actually **lose money** overall. Even though millions of customers would respond, the campaign's average response rate would still fall short of the 9.7% breakeven threshold, resulting in a small but negative return.

Question 5: RFM-Based Targeting

By applying the breakeven response rate as a cutoff, DoorDash can focus only on RFM segments where the **average response rate exceeds 9.7%**.

- **A) Targeted customers:** 15,679,950
- **B) Expected responders:** 2,015,323
- **C) Projected total profit:** $19,949,523.51
    - Marketing spend: $15,052,752.41
    - Net projected profit: **$4,896,771.09**
    - **ROMI: 0.33 (33%)**

This selective targeting strategy yields a **positive return on marketing**. Although DoorDash contacts fewer customers, it earns nearly $5 million in net profit, demonstrating that smarter targeting outperforms blanket promotion.

Question 6: Comparing "Send to Everyone" vs. RFM Targeting

Comparing both strategies reveals a significant improvement:

Strategy	Net Profit	ROMI
Send to Everyone	–$1.50M	–0.04 (-4%)
RFM Targeting	+$4.90M	+0.33 (33%)

Targeting based on RFM segmentation not only saves on marketing costs but also increases profitability by about **$6.4 million**. The improvement arises because DoorDash avoids sending notifications to low-response customers who would likely ignore or mute the app. In essence, **precision marketing replaces mass marketing**, achieving higher efficiency with fewer messages.

Question 7: Limitations of the Marginal Cost Calculation

Michelle's team calculated the marginal cost of a push notification as:

$$\text{Marginal Cost} = \text{Opt-out Probability} \times \text{Lifetime Value Gap} = 0.03 \times \$32 = \$0.96$$

While this provides a useful starting point, it has several limitations:

1. **Assumed Linear Effect:** The model assumes that each new notification increases opt-outs by a constant 3%, which might not hold true—fatigue effects can accelerate over time.
2. **Ignoring Customer Heterogeneity:** The calculation treats all users as equally likely to opt out, even though some may tolerate more messages than others.
3. **Static LTV Estimate:** It uses a fixed $32 lifetime value difference, but LTV likely changes with time, market conditions, or user behavior.
4. **No Control for External Factors:** External promotions, seasonality, or competitor activity could influence opt-out rates independently of notification frequency.

To improve accuracy, DoorDash could run **controlled A/B experiments** to measure incremental opt-outs and LTV impacts over different messaging frequencies and customer segments. Incorporating these dynamic effects would make the marginal cost estimate more realistic.

Question 8: Caveats and Case Limitations

Several caveats should be considered when interpreting this analysis:

- **Simplified Profit Assumptions:** The case assumes that the "profit" column fully captures all costs, but in reality, marketing and operational costs can vary by order type or region.
- **Short-Term Campaign Data:** The test measures only a one-week response window, which may not capture delayed orders or long-term engagement effects.
- **Uniform Scaling:** The projection to the full customer base assumes that test-sample behavior generalizes perfectly to all users, which may not be realistic.
- **No Incrementality Control:** The analysis cannot distinguish between habitual orders and truly incremental orders caused by the promotion.
- **Behavioral Assumptions:** It assumes customers' responsiveness will remain constant across campaigns, even though over-messaging could reduce effectiveness over time.

Overall, while the RFM and breakeven analyses provide clear managerial insights, they simplify complex customer behavior. In practice, DoorDash should complement these findings with **longitudinal experiments** and **incrementality testing** before deploying large-scale promotions.

Summary Insight:
RFM-based targeting clearly delivers stronger profitability and a positive ROMI, showing that **precision marketing built on customer behavior data** can balance growth and cost efficiency, exactly the challenge Michelle Torres set out to solve.